

Amal Nath VJ

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Experience

UVJ Technology | Software Engineer(*Cold-chain logistics / Physics-informed ML*)

Trivandrum

- Engineered TempTrack AI, a full-lifecycle pharmaceutical cold-chain platform combining physics-based thermal simulation, dual-model ML risk prediction, and automated document intelligence to flag at-risk shipments before and during transit.
- Built a Random Forest pre-shipment failure classifier (24 h before departure) with a leakage-aware two-tier feature design :- Level 0 raw shipment inputs and Level 1 derived route-climate features ,backed by a continuous-learning retraining loop that ingests real post-delivery outcomes and gates production promotion behind a ROC-AUC previous check, with auto-backup of replaced models.
- Built a Fourier's Law-based thermal simulation engine that ingests real QC-report container parameters and predicts per-shipment time-to-failure projecting the exact hour at which the cargo would breach its safe temperature threshold under the planned route's ambient conditions.
- Built a multi-stage AI extraction pipeline (Claude text+vision for QC reports, autonomous Claude tool-use agent for manufacturer-site web search) with a 3-layer validator :- range guards, cross-field consistency checks, and a Fourier physics back-check that re-simulates extracted parameters and flags >50 percentage deviation from the report's qualified duration , catching LLM hallucinations before they corrupt the simulation.

Siemens Technology and Services Pvt. Ltd | Intern

Chennai

- Worked on smart building projects involving access controllers (Sipass), SDT (System Design Tool), and surveillance systems.

Education

College of Engineering Trivandrum

2021 - 2025

B.Tech in Computer Science and Engineering

Projects

Credit Card Risk Prediction 🧠

Machine Learning, XGBoost ,Pandas , Numpy

- Built an end-to-end credit card risk model to assess default risk on loan, customer, and bureau data, consolidating multiple sources and performing EDA + feature engineering.
- Designed, tuned, and compared Logistic Regression, Random Forest, and XGBoost models; addressed class imbalance and improved discrimination as measured by ROC-AUC, KS, and Gini.

Flipkart E-Commerce Chatbot 🧠 Python, Streamlit, Groq, LLaMA 3.3, ChromaDB, Semantic Router, SQLite

- Built an end-to-end AI chatbot for an e-commerce platform that handles both FAQ queries and real-time product search using intent-based routing with Semantic Router.
- Implemented a RAG pipeline using ChromaDB and Sentence Transformers to retrieve relevant FAQ answers, and a Text-to-SQL pipeline using Groq (LLaMA 3.3) to query a SQLite product database based on natural language input.

Log Classification with Hybrid Framework 🧠

NLP , LLM ,ML, FastAPI

- Built an intelligent log classification system for production system logs (errors, warnings, notifications) to reduce manual triage.
- Implemented a 3-tier hybrid pipeline (Regex, Sentence-Transformer + Logistic Regression, LLM with confidence-based fallback) to handle both structured and noisy legacy logs.
- Boosted classification accuracy by 38–42% and cut manual log triage effort by 30%.

Car Crash Damage Classification 🧠

DeepLearning , PyTorch, ResNet50

- Built a deep learning model to classify car damage from images into 6 categories (front/rear: breakage, crushed, normal) to support faster, consistent claim assessment.
- Fine-tuned a ResNet50 backbone with transfer learning and deployed a Streamlit app so users can upload images and get real-time damage predictions.

Technical Skills

Languages: Python, SQL

ML / DL Frameworks: PyTorch, Scikit-learn, XGBoost, HuggingFace Transformers, spaCy

NLP and LLMs: Sentence Transformers, LangChain, LLaMA, Groq API, Claude API, RAG, Semantic Router, ChromaDB

Data Science: NumPy, Pandas, Matplotlib, Seaborn, Feature Engineering, EDA

Backend / Databases: FastAPI, PostgreSQL, SQLAlchemy, Streamlit

Developer Tools: Git, GitHub

Domains: Machine Learning, Deep Learning, NLP, Computer Vision, Large Language Models, GenAI, Agentic AI, Physics-informed ML

Achievements & Leadership

Siemens Scholar (2021–2025): Selected among top engineering students in India based on academic merit.

Leader, CET Official Dance Crew (2021–2025): Led the team to the finals of the Destiny Dance Competition, representing Kerala at the national level.